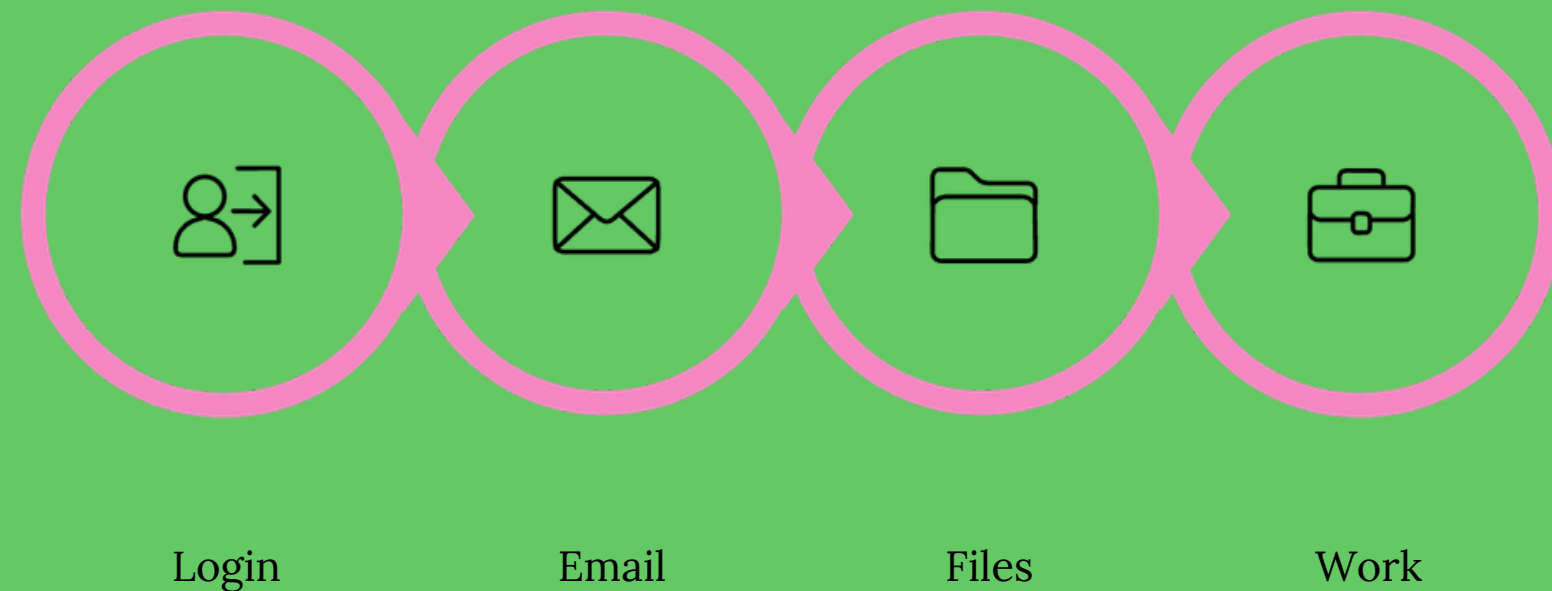


CyberTrace

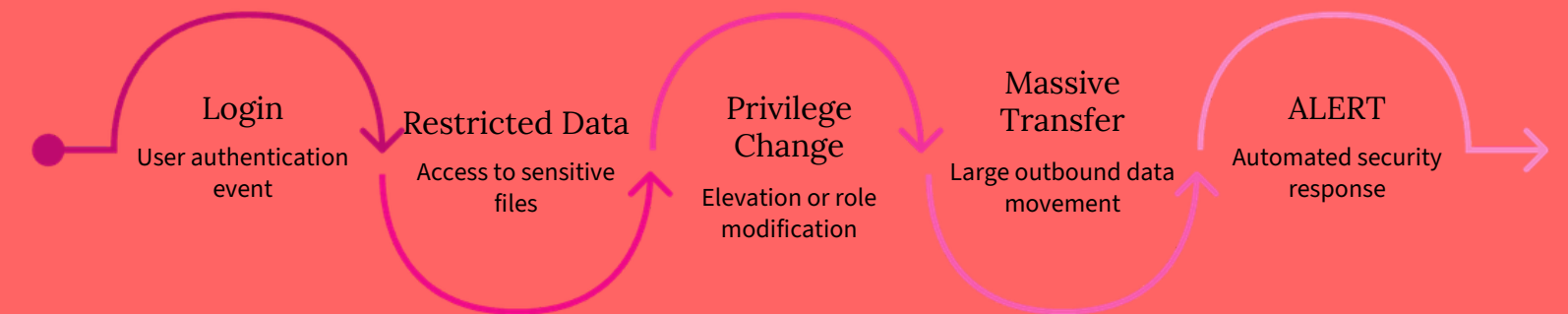
ML-Based User Behaviour Analysis for Multi-Stage Cyberattack Detection

From detecting individual events to understanding the behaviour behind them.

✓ NORMAL ACTIVITY



🚨 SUSPICIOUS PATTERN



📄 CyberTrace analyzes **sequences of activities**, not individual events — revealing the story behind the behaviour.

The Problem: One Suspicious Action Doesn't Tell the Whole Story

Modern cyberattacks unfold across multiple coordinated steps. Evaluating each event in isolation misses the bigger picture — and leaves security teams overwhelmed.

✓ NORMAL EMPLOYEE ACTIVITY

01

Login

02

Check Emails

03

Access Project Files

04

Work → Logout

🚨 SUSPICIOUS BEHAVIOUR SEQUENCE

01

Login

02

Restricted Files

03

Privilege Escalation

04

Sensitive DB Access

05

Large Data Transfer

Sequences reveal intent

A single login is normal — the sequence reveals the story

Multi-step attacks

Modern attacks involve multiple coordinated steps that appear benign individually

Alert fatigue

Security teams drown in thousands of daily alerts without behavioural context

Context is critical

Connecting related activities provides the attack context needed for investigation

Existing Approach vs CyberTrace

The fundamental difference is not just what is detected — it is **how** the analysis is framed. Existing tools ask whether an event is suspicious. CyberTrace asks whether a **sequence of activities** looks suspicious.

⚠️ EXISTING SYSTEMS

Event-level detection

Focus on individual suspicious events or known attack signatures

Disconnected alerts

Multiple events appear as separate, unrelated alerts — e.g. Unusual Login → 🚨 Alert

Manual correlation

Analyst must manually connect the dots across hundreds of alerts

Limited context

No behavioural baseline — cannot distinguish deviation from normal patterns

● CYBERTRACE

Sequence analysis

Analyzes sequences of user activities across time, not isolated events

Behavioural baseline

Learns normal behavioural patterns per user and detects significant deviations

Event correlation

Connects related events into coherent behavioural patterns automatically

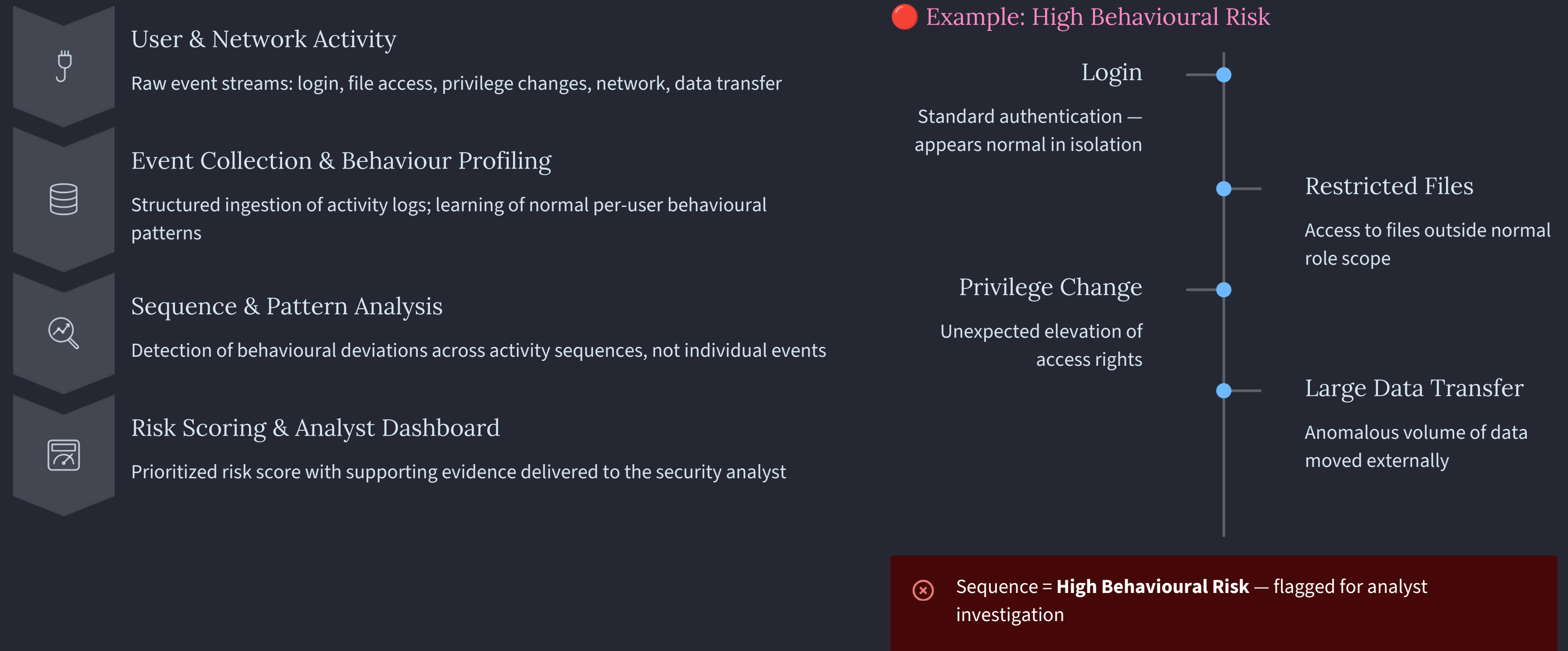
Risk score + evidence

Generates a risk score with supporting evidence for analyst review

📌 **Key Difference:** Existing: *"Is this event suspicious?"* → CyberTrace: *"Does this **sequence of activities** look suspicious?"*

How CyberTrace Works

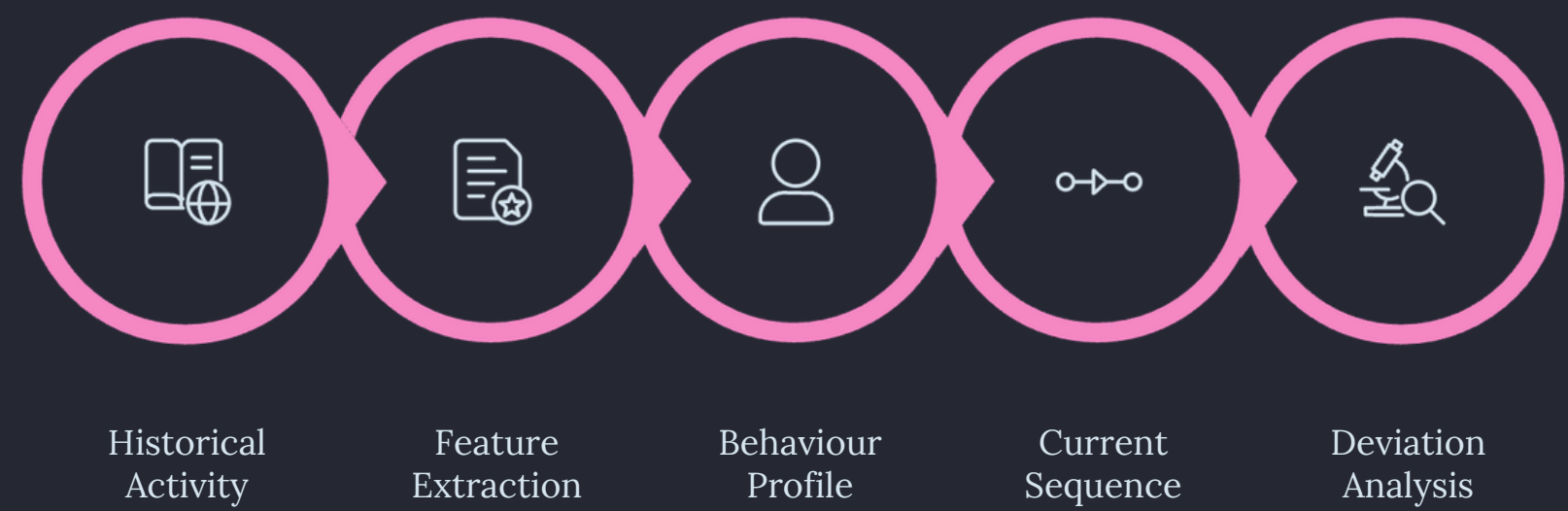
CyberTrace processes raw user and network activity through a structured pipeline — transforming millions of events into prioritized, evidence-backed behavioural risk assessments.



Machine Learning Approach

CyberTrace combines multiple ML techniques to build per-user behavioural baselines and evaluate current activity sequences against them — producing a risk score grounded in evidence, not rules.

Conceptual Architecture



🔑 Key Innovation

Instead of evaluating events independently, CyberTrace **combines multiple behavioural signals and their sequence** to identify suspicious activity patterns.

Scoring Factors

Baseline deviation	Event sequence
Unusual frequency	Event timing
Resource sensitivity	Activity relationships

⚠️ **Important:** CyberTrace identifies and prioritizes *potentially malicious behaviour* for investigation — it does **not** prove an attack occurred or identify a specific actor.



Anomaly Detection

Flags activity that deviates significantly from the established per-user behavioural baseline



Clustering

Groups users with similar behavioural profiles to improve baseline accuracy and reduce false positives



Sequence Analysis

Evaluates the order and relationship of events — the core differentiator of the CyberTrace approach

Who Benefits & Why It Matters

Primary Users

SOC Analysts & Cybersecurity Teams

- Reduced manual effort
Automatically connects thousands of security events into coherent behavioural patterns
- Automatic prioritization
Suspicious behavioural patterns surfaced first — not buried in alert queues
- Context over alerts
Provides investigative context instead of isolated, disconnected notifications
- Faster investigation
Accelerates response to potential multi-stage attacks across enterprises, banks, hospitals, and cloud environments

Impact Comparison

✗ Without CyberTrace

Millions of Events → Many Disconnected Alerts → **Manual Investigation** → Attack Eventually Identified

✓ With CyberTrace

Millions of Events → Behaviour Analysis → Related Events Connected → **Suspicious Patterns Prioritized** → Faster Investigation

1

Analyst Enabler

CyberTrace does not replace the security analyst — it helps them understand **which behaviour deserves attention** and **why**

3x

Signals Combined

Sequence + deviation + evidence — three layers of context per risk score

Detect behaviour. Connect events. Explain risk.